nature portfolio

Peer Review File

Integration of fairness-awareness into clinical language processing models

Corresponding Author: Dr Ervin Sejdic

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
The study presents a novel fairness-aware NLP model for race extraction and classification from EHR notes. An active learning approach was used to curate a representative dataset. The experiments demonstrated that the CNN model outperformed transformer models in both accuracy and fairness. The authors also identified persistent performance disparities by sex and age. Overall, the study addresses a highly significant problem. The analysis is comprehensive, supported by solid statistical methods and clear scientific visualizations. Another strength is the well-articulated adoption of process-oriented fairness constraint methods. I also found the discussion is very throughtful. However, I have some comments and concerns regarding the methodology and study design. Please see my detailed feedback below.

INTRO/Related work
Would be good to consider refer previous studies that has estimated the rate of missing of race information in different structured EHRs. For example, Sholle et al reported using NLP identified 948 additional patients as black, a 26%increase, and 665 additional patients as Hispanic, a 20% increase (Sholle 2019).

Sholle ET, Pinheiro LC, Adekkanattu P, Davila MA, Johnson SB, Pathak J, Sinha S, Li C, Lubansky SA, Safford MM, Campion TR. Underserved populations with missing race ethnicity data differ significantly from those with structured race/ethnicity documentation. Journal of the American Medical Informatics Association. 2019 Aug 1;26(8-9):722-9.

Furthermore, in addition to the transformer-based approach, the authors should also mention rule-based/symbolic methods or symbolic + transformer-based hybrid architectures. It is known that rule-based models suffer from challenges in handling variant linguistic and syntactic expressions. However, from the perspective of improving the overall documentation of race information in the entire EHR, one key use case for the proposed model is to screen all EHR notes and populate race information to enhance structured demographic fields. Given that a single institution’s clinical documentation may contain over 300 million notes, with an average token length of 2,500 tokens per note based on MIMIC data, this results in substantial computational and financial burdens. For example, using eight H100 GPUs to run RoBERTa and assuming an optimistic estimate of 0.5 seconds per document, full cohort screening would still take more than half a year. Therefore, it would be good to enhance the related work section, particularly the rationale for selecting this approach and how it ties to the application use case. In the discussion section, I would also encourage the authors to discuss the strengths and limitations of the proposed approach from the perspective of real-world use case scenarios.

Study objective: please explicit define the task more clearly. There are many ways for race information extraction ranging from traditional NER or clinical concept extraction to machine classification. Within classification task, you can define from sentence level, paragraph level to document level. Each task objective link with different usecase and motication. Please justify 1) detail task definition and 2) why selecting this approach.

In addition, please re-order the four study objectives based on the nature order e.g. active learning should occur first

RESULES

METHODS/RESULTS

Active learning pipeline
Did we assess the bias and fairness of the BERT-based classification model used in the active learning pipeline? Assuming the model is not entirely free of bias, could there be a risk of unequally distributed false negative errors among underrepresented groups, particularly those with sparse data and poor data quality? In this sense, the training sample may inherit potential biases that could affect the development of downstream models.

Data preprocessing
I have some concerns regarding the data preprocessing procedures of comparing the proposed model with the BERT-based baseline models. For the BERT-based models, the authors state that "each document was treated as a single flat sequence and tokenized using their corresponding pre-trained tokenizer. Truncation and padding were applied to a fixed maximum input length of 512 tokens, in accordance with model-specific constraints." In contrast, for the proposed model, the authors applied sentence segmentation and used sentences as the unit of input, preserving more of the document structure. This introduces a potential bias in the comparison. It is well known that BERT-based models struggle with long document classification tasks due to the 512-token input limitation. The methods between directly hard cut by 512 token vesus applying a sentence detection algorithm first can make big difference in presenting the appropriate contextual information. Therefore, treating each document as a flat sequence without structural decomposition is arguably not the most suitable preprocessing strategy for these models. This discrepancy in input preprocessing likely contributes to the performance gap reported in Table 1 and raises concerns about the validity of the comparison—it is not an apples-to-apples evaluation. A more appropriate experimental design would involve either (1) using the same input preprocessing across all models, or (2) optimizing the preprocessing strategy for each model type while clearly justifying the rationale and addressing the implications for comparison fairness. The bias resulted in the flat sequence methods also partically explained in Table 1 and Figure 2, where macro f1-score for BERT variants model are really bad and particularly for BERT-base, RoBERT and DeBERT with some training convergence issues. Event the class label is imbalance and race, I am very surprived about the BERT model (even without fairness adjustment) cannot make correct classification for other race. Previous studies (cited below) that used BERT reported varient performence (0.6-0.8) for classifying other race categories.

Don’t Walk OJ, Pichon A, Nieva HR, Sun T, Altosaar J, Natarajan K, Perotte A, Tarczy-Hornoch P, Demner-Fushman D, Elhadad N. Auditing Learned Associations in Deep Learning Approaches to Extract Race and Ethnicity from Clinical Text. InAMIA Annual Symposium Proceedings 2024 Jan 11 (Vol. 2023, p. 289).
Parasurama P. raceBERT--A Transformer-based Model for Predicting Race and Ethnicity from Names. arXiv preprint arXiv:2112.03807. 2021 Dec 7.

Training and hyperparameter tuning
Author mentioned all models were trained and evaluated using 10-fold stratified cross-validation and then 10% of the annotated dataset was held out using a stratified shuffle split for final performance evaluation. The result Tables showed "10-fold cross-validation", it is unclear to me whether the reported performance is based on cross-validation set or held out set. Please clarify if you use cross-validation or nested-cross-validation design.

Code availability and open source
It is very important to release the model source code to ensure reproducibility.

DISCUSSION
Fairness-aware BERT already performed reasonable well. Given this strong baseline, do you think it may be more impactful to further optimize or adapt the existing BERT architecture—such as through fairness-regularized fine-tuning, domain-specific pretraining

Minor
The term EHR (electronic health records) is more commonly used than EMR by the community

Reviewer #2

(Remarks to the Author)
Brief Summary: The paper discusses the importance of fairness in training medical machine learning models but states that fairness can not always evaluated or optimized for due to the typical lack of structured race (or other demographic) labels often found with electronic medical records. They propose to fix this problem by training classifiers to perform race detection and annotation based on EMRs (in this case text reports). They then perform comprehensive tests on a series of models, mostly BERT-style LLMs and then a novel hierarchical convolution model aimed to mirror the hierarchical structure of medical records (with patients having multiple visits with multiple reports with multiple sentences with multiple words). They find that the hierarchical convolution model performs best and their attempts at fairness aware training largely do not work.

Overall Impression: This is undoubtedly an important topic and I believe the paper was well-written both in contextualizing the work and covering the experiments and their results. The experiments were thorough and covered a variety of different angles of discussion. However, I struggled to come away with a single valuable takeaway message beyond that the classification of race is a feasible (and maybe easy given the extremely high performance metrics) task with the hierarchical convolution network being a good approach at doing so. I wonder if a more meaningful performance delta (the original LLM results are so poor as to be not valuable as a comparison, and then the fairness-aware results largely reverse the trend with BERT outperforming the CNN-based approach) or a downstream application (e.g. showing labels from this classification method can improve the fairness aware training or be used for evaluation of other downstream models) would be valuable in telling a more meaningful story with this work.

Specific Comments/Questions: (with recommendations for addressing each comment)
1. The authors mention that only a few papers studied this topic, and only two evaluate bias in the extraction of social constructs from EMRs, but those studies were not then contrasted with such that it wasn't clear what the advantage or continuation of progress over those papers is.
2. The BERT-based LLM's perform so poorly on what seems to be a relatively simple tasks (as noted by the hierarchical CNN performing incredibly well) that it causes worries of some problem during training, especially when the fairness aware training largely fixes the problem (and not just at the expense of majority group performance).
3. This then mitigates a central point of the discussion of worrying about LLMs detecting and propagating hidden biases as the results seemingly show they can't detect them. Addressing point 2 would fix this, but as it stands that discussion point felt hollow.
4. The Hierarchical CNN does't seem to have 512 max token length (or it does but just per sentence which are all handled differently)? Could this be a source of the performance gap? Additional results even just in the supplement regarding the rate of truncation, analysis to see whether and how often the label information was lost through truncation, performance on truncated vs. non-truncated data points would be valuable.
5. I worry given the small size of the dataset that intersectional biases may be extremely noisy, especially for minority groups given their small prevalence and the large number of groups to look for bias in.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
Thank you for the detailed response to my comments. My previous concerns regarding data pre-processing and evaluation design has been address adequately. One minor point is I couldn't find any Github/huggingface link, I would strongly encourage the author to release the source code for model architecture (not the trained model) to promote open science.

Reviewer #2

(Remarks to the Author)
I appreciate the thoroughness of the author response to each reviewer point. The added detail with respect to both the framing and the experimentation were comprehensive and valuable. The only remaining point I would like to follow-up on is the length of the reports, specifically with respect to number of tokens given a BERT window size of 512, which both reviewers noted. The response that most reports were short and that is a satisfactory answer to the main worry of experimental unfairness. However, it does raise two follow ups that I wanted to clarify.
1. Are there any examples of longer data points and how the model does there, even if it is just a case study. These short reports are somewhat out of distribution from my expectation of automatic EHR document labeling, and while that is not an issue it does make me wonder about the performance in a long or multi-document setting, which is common in the healthcare domain.
2. Following up on Reviewer 1's discussion of the fundamental approach/task difference between BERT and the author's proposed sentence-by-sentence convolutional approach (which I felt the author's response to was valid and convincing), I wonder if a helpful middle ground that could (hopefully easily) be added as a baseline would be a sentence by sentence BERT embedding with some simple aggregation (either mean prediction or max prediction to better isolate the information). While not as standard a baseline, I do wonder if this could help to highlight fairness, loss, and other training contributions beyond the more granular sentence-by-sentence breakdown, especially given the relative lack of ablation baselines.

Version 2:

Reviewer comments:

Reviewer #2

(Remarks to the Author)
Thank you for addressing all remaining points. I am satisfied with the most recent version of the manuscript.

Responses to Editorial Decision Letter on the paper titled:

Rethinking clinical NLP: Hierarchical architectures, fairness-aware models, and active annotation for race detection

Manuscript# COMMSMED-25-1517-T

Thank you for giving us the opportunity to resubmit a revised manuscript. We have provided detailed point-to-point responses according to the comments raised by the reviewers below. Furthermore, we have revised the manuscript in which all revisions are made with a blue colored text.

Response to Reviewer #1's Comments:

Reviewer #1's comment 1:

Would be good to consider refer previous studies that has estimated the rate of missing of race information in different structured EHRs. For example, Sholle et al reported using NLP identified 948 additional patients as black, a 26%increase, and 665 additional patients as Hispanic, a 20% increase (Sholle 2019).

Sholle ET, Pinheiro LC, Adekkanattu P, Davila MA, Johnson SB, Pathak J, Sinha S, Li C, Lubansky SA, Safford MM, Campion TR. Underserved populations with missing race ethnicity data differ significantly from those with structured race/ethnicity documentation. Journal of the American Medical Informatics Association. 2019 Aug 1;26(8-9):722-9.

Authors' response 1:

We thank the reviewer for this important suggestion. We have expanded our discussion of missing race/ethnicity data in structured EHR fields by adding the Sholle et al. (2019) reference along with additional supporting literature. Specifically, we have revised the second paragraph of the Introduction to include these references and provide specific quantitative evidence of data missingness in structured EHR fields.

"Several studies have quantified the extent of missing or incomplete race and ethnicity information within structured EHR fields, revealing that these gaps remain a significant barrier to equitable data representation. Analyses of clinical text have revealed an additional 948 patients identified as Black (+26%) and 665 as Hispanic (+20%) beyond what was captured in structured EHR fields[13]. Across EHRs, missing or uninformative race entries range from 25% to over 57%[14], with misclassification disproportionately affecting multiracial patients and other minoritized groups[15]. These omissions are not merely technical deficiencies; they distort population denominators, obscure health inequities, and introduce systematic bias into downstream analyses"

13. Sholle, E. T. et al. Underserved populations with missing race ethnicity data differ significantly from those with structured race/ethnicity documentation. J. Am. Med. Informatics Assoc. 26, 722–729 (2019).

14. Polubriaginof, F. C. et al. Challenges with quality of race and ethnicity data in observational databases. J. Am. Med. Informatics Assoc. 26, 730–736 (2019).

15. Proumen, R., Connolly, H., Debick, N. A. & Hopkins, R. Assessing the accuracy of electronic health record gender identity and REal data at an academic medical center. BMC Heal. Serv. Res. 23, 884 (2023).

Reviewer #1’s comment 2:

Furthermore, in addition to the transformer-based approach, the authors should also mention rule-based/symbolic methods or symbolic + transformer-based hybrid architectures. It is known that rule-based suffer from challenges in handling variant linguistic and syntactic expressions. However, from the perspective of improving the overall documentation of race information in the entire EHR, one key use case for the proposed is to screen all EHR notes and populate race information to enhance structured demographic fields. Given that a single institution’s clinical documentation may contain over 300 million notes, with an average token length of 2,500 tokens per note based on MIMIC data, this results in substantial computational and financial burdens. For example, using eight H100 GPUs to run RoBERTa and assuming an optimistic estimate of 0.5 seconds per document, full cohort screening would still take more than half a year. Therefore, it would be good to enhance the related work section, particularly the rationale for selecting this approach and how it ties to the application use case. In the discussion section, I would also encourage the authors to discuss the strengths and limitations of the proposed approach from the perspective of real-world use case scenarios.

Authors' response 2:

We thank the reviewer for this important observation regarding rule-based/hybrid methods and computational considerations for large-scale deployment. We have revised the manuscript to address these concerns comprehensively.

<u>Specific changes made:</u>

- Added a new paragraph discussing rule-based and hybrid approaches in the **Introduction**, their computational advantages, and the trade-offs between different architectures. We now explicitly state that this study focuses on research applications where accuracy and fairness detection are prioritized over institutional-scale deployment efficiency.

- Added a new paragraph in the **Discussion** that directly addresses the reviewer's concern about computational feasibility for processing millions of notes. We acknowledge that: (1) Large-scale EHR screening requires computationally efficient approaches; (2) Rule-based or hybrid methods are more practical for institutional-scale demographic field population; (3) Our fairness-aware models are designed for research, bias auditing, and equity evaluation contexts; (4) A tiered approach (efficient screening then fairness-aware models for uncertain/high-stakes cases) offers a pragmatic compromise.

We agree that for populating structured fields across hundreds of millions of notes, hybrid approaches are more practical. However, our models serve a complementary purpose: ensuring equitable outcomes in research and high-stakes decisions where the costs of biased predictions outweigh computational expense. The manuscript now clarifies this distinction and provides guidance on when each approach is most appropriate.

Reviewer #1’s comment 3:

Study objective: please explicit define the task more clearly. There are many ways for race information extraction ranging from traditional NER or clinical concept extraction to machine classification. Within classification task, you can define from sentence level, paragraph level to document level. Each task objective link with different usecase and motication. Please justify 1) detail task definition and 2) why selecting this approach.

In addition, please re-order the four study objectives based on the nature order e.g. active learning should occur first

Authors' response 3:

We thank the reviewer for this constructive feedback. The study objectives paragraph has been substantially revised to explicitly define the task as *document-level race classification* rather than sentence- or token-level extraction, and to justify this choice as aligned with the use case of augmenting structured demographic fields. The sequence of objectives has also been reordered to follow their natural methodological flow (beginning with active learning, followed by model comparison, fairness evaluation, and bias analysis). This revision clarifies both the conceptual framing and the practical rationale of the study. Here is the changed paragraph:

"This study investigates the automatic document-level classification of patient race from unstructured clinical text within EMRs as a critical step toward capturing social constructs that inform care phenotypes, enabling the assessment of healthcare equity and the identification of populations receiving lower-quality care. Unlike named-entity recognition or sentence-level concept extraction, which identify local mentions of race within notes, our objective is to determine each patient's overall racial category based on their complete longitudinal documentation. This formulation aligns with the practical use case of augmenting or validating structured demographic fields and enables population-level analyses of equity and care quality. To achieve this, we first implemented an active learning framework to efficiently guide the annotation process by prioritizing notes likely to contain race-related information. We then compared state-of-the-art transformer-based models with a hierarchical convolutional neural network designed to represent the multi-level structure of clinical documentation, thereby capturing contextual dependencies across sentences and encounters. Next, we evaluated the impact of fairness-aware optimization techniques, including a loss function inspired by equalized odds, to determine whether such constraints improved predictive balance across racial groups. Finally, we analyzed potential sources of bias by examining how dataset composition and sensitive attributes such as sex and age influenced model performance. This work seeks to develop more robust, interpretable, and equitable methods for racial, and more widely social determinants of health, information extraction from clinical text, ultimately supporting the creation of more representative datasets for health equity research."

Reviewer #1's comment 4:

Active learning pipeline
Did we assess the bias and fairness of the BERT-based classification used in the active learning pipeline? Assuming the is not entirely free of bias, could there be a risk of unequally distributed false negative errors among underrepresented groups, particularly those with sparse data and poor data quality? In this sense, the training sample may inherit potential biases that could affect the development of downstream models.

Authors' response 4:

We thank the reviewer for this insightful comment. We agree that the fairness of the model guiding active learning represents an important and often overlooked source of potential bias propagation. To address this, we have clarified in the **Methods (Active Learning)** section that the BERT-base model used to guide sample selection was intentionally trained without fairness constraints to evaluate whether conventional uncertainty-based active learning, which is commonly used in clinical NLP, could introduce demographic bias into the annotated dataset. We also added a statement noting that the resulting distributional analysis (reported in the **Results**) showed minimal but non-negligible demographic shifts between the baseline and actively sampled subsets, suggesting subtle sampling bias.

In the **Discussion** section, we expanded our analysis to explicitly acknowledge that active learning itself can amplify representational disparities when the query model's uncertainty is correlated with data density or group frequency. We now discuss this as a broader issue in the literature, citing recent work highlighting the lack of fairness-aware sampling approaches in clinical NLP. This addition emphasizes that fairness considerations should extend beyond predictive modeling to the entire data pipeline (including data selection) to prevent the amplification of representational bias in labeled datasets.

We would also like to clarify that all labels used in this study were assigned by human annotators. The BERT-based model was used solely to prioritize which notes were more likely to contain race information, not to automatically label them. Therefore, model errors could only influence which notes were selected for review (sampling bias), not the correctness of the labels themselves. We explicitly examined this potential sampling bias by comparing demographic distributions before and after active learning, and observed minimal but non-negligible shifts.

These revisions clarify our rationale, demonstrate awareness of potential bias in the active learning process, and strengthen the fairness framing of the manuscript.

Reviewer #1's comment 5:

Data preprocessing
I have some concerns regarding the data preprocessing procedures of comparing the proposed with the BERT-based baseline models. For the BERT-based models, the authors state that "each document was treated as a single flat sequence and tokenized using their corresponding pre-trained tokenizer. Truncation and padding were applied to a fixed maximum input length of 512 tokens, in accordance with model-specific constraints." In contrast, for the proposed model, the authors applied sentence segmentation and used sentences as the unit of input, preserving more of the document structure. This introduces a potential bias in the comparison. It is well known that BERT-based struggle with long document classification tasks due to the 512-token input limitation. The methods between directly hard cut by 512 token vesus applying a sentence detection algorithm first can make big difference in presenting the appropriate contextual information. Therefore, treating each document as a flat sequence without structural decomposition is arguably not the most suitable preprocessing strategy for these models. This discrepancy in input preprocessing likely contributes to the performance gap reported in Table 1 and raises concerns about the validity of the comparison—it is not an apples-to-apples evaluation. A more appropriate experimental design would involve either (1) using the same input preprocessing across all models, or (2) optimizing the preprocessing strategy for each type while clearly justifying the rationale and addressing

the implications for comparison fairness. The bias resulted in the flat sequence methods also partically explained in Table 1 and Figure 2, where macro f1-score for BERT variants are really bad and particularly for BERT-base, RoBERT and DeBERT with some training convergence issues. Event the class label is imbalance and race, I am very surprived about the BERT (even without fairness adjustment) cannot make correct classification for other race. Previous studies (cited below) that used BERT reported varient performence (0.6-0.8) for classifying other race categories.

Don't Walk OJ, Pichon A, Nieva HR, Sun T, Altosaar J, Natarajan K, Perotte A, Tarczy-Hornoch P, Demner-Fushman D, Elhadad N. Auditing Learned Associations in Deep Learning Approaches to Extract Race and Ethnicity from Clinical Text. InAMIA Annual Symposium Proceedings 2024 Jan 11 (Vol. 2023, p. 289).
Parasurama P. raceBERT--A Transformer-based for Predicting Race and Ethnicity from Names. arXiv preprint arXiv:2112.03807. 2021 Dec 7.

Authors' response 5:

We thank the reviewer for this important methodological concern about preprocessing strategies and their potential impact on fair model comparison. Our corpus consists of the social history and risk factor sections from primary care EMRs, which are substantially shorter than full clinical narratives. Analysis revealed that 98% of documents fit within 512 tokens. Therefore, the 512-token truncation did not create information availability disparity between models, both architectures had access to complete document content.

The sentence segmentation for the hierarchical CNN is an architectural requirement, not a discretionary choice. The model requires sentence boundaries to compute word-level and sentence-level attention. Applying sentence segmentation to BERT would require substantial architectural modifications (e.g., adding sentence encoders), effectively creating a different model architecture. Our goal was to compare architectures as they are typically deployed in practice. Each model used its recommended preprocessing pipeline from the literature: flat sequence processing for transformers and hierarchical segmentation for the CNN. This approach reflects how these models would be implemented in real-world applications.

The reviewer cites studies reporting F1 scores of 0.6-0.8 for BERT-based race classification. However, those studies either addressed binary classification tasks (e.g., Black vs. non-Black) (Don't Walk OJ, et al.) or simpler multi-class problems using clean, highly lexical inputs such as names where large, balanced datasets (over 13 million entries) and explicit cues yield artificially high performance (Parasurama P, 2021). Whereas our task involves nine-class classification with extreme class imbalance (76% negative class, nine positive classes). Our BERT achieved 84.38% micro-precision but struggled with minority class generalization (66.84% macro-F1), reflecting the known challenge of transformers on highly imbalanced multi-class problems.

We have clarified in the Methods section that: (1) each model used its standard preprocessing pipeline appropriate to its architecture, (2) our documents are substantially shorter than typical clinical notes and fit within the 512-token window, ensuring fair information access across models, and (3) the hierarchical CNN's sentence-based approach is an architectural requirement rather than a modifiable preprocessing choice.

Reviewer #1's comment 6:

Training and hyperparameter tuning
Author mentioned all were trained and evaluated using 10-fold stratified cross-validation and then 10% of the annotated dataset was held out using a stratified shuffle split for final performance evaluation. The result Tables showed "10-fold cross-validation", it is unclear to me whether the reported performance is based on cross-validation set or held out set. Please clarify if you use cross-validation or nested-cross-validation design.

Authors' response 6:

We thank the reviewer for requesting clarification on our evaluation design. We apologize for any confusion in the original presentation. We used a hold-out validation design with stratified sampling where we first reserved 10% of the annotated dataset as an independent test set using stratified shuffle split to maintain class proportions. This test set was never used for model training, hyperparameter tuning, or model selection then the remaining 90% was used for 10-fold stratified cross-validation. We have clarified this design in the **Training and hyperparameter tuning** subsection and the tables that use 10-fold cross validation in the **Results**.

Reviewer #1's comment 7:

Code availability and open source
It is very important to release the source code to ensure reproducibility.

Authors' response 7:

We agree with the reviewer that code availability is essential for reproducibility. In response to this concern, we have added a comprehensive **Code availability** subsection to the manuscript's **Method's** section.

Reviewer #1's comment 8:

DISCUSSION
Fairness-aware BERT already performed reasonable well. Given this strong baseline, do you think it may be more impactful to further optimize or adapt the existing BERT architecture—such as through fairness-regularized fine-tuning, domain-specific pretraining

Authors' response 8:

We thank the reviewer for this thoughtful observation. The reviewer is correct that fairness-aware BERT achieved strong performance (macro-F1 = 91.48%), and we agree that further optimization of transformer architectures through fairness-regularized fine-tuning, domain-specific pretraining, or architectural modifications represents a valuable research direction.

However, our study was designed to challenge a different question: *whether the default reliance on transformer architectures is always optimal for clinical NLP tasks*, particularly when dealing with structured, hierarchical clinical documentation and extreme class imbalance. We believe both approaches are important and complementary:

We agree with the reviewer that optimizing existing BERT architectures is valuable, particularly given their widespread adoption. However, we believe our contribution is complementary: demonstrating that **task-**

specific architectural design remains relevant in the transformer era, especially for applications requiring interpretability, fairness, and explicit modeling of document structure. The field benefits from both approaches, optimizing popular architectures AND exploring whether alternative designs better match specific data characteristics. While transformer optimization is already extensively explored in the literature, our work contributes to the less-studied question of when task-specific architectures may be more appropriate than adapting transformers.

We appreciate the reviewer's suggestion and agree that future work could explore whether similar architectural principles (fairness regularization, etc.) could be integrated into transformer-based models to achieve the best of both approaches.

Reviewer #1's comment 9:

Minor
The term EHR (electronic health records) is more commonly used than EMR by the community

Authors' response 9:

We thank the reviewer for this observation. We have updated the manuscript to use "electronic health record (EHR)" consistently throughout, replacing all instances of "EMR" to align with standard community terminology.

Response to Reviewer #2s Comments:

Reviewer #2's comment 1:

1. The authors mention that only a few papers studied this topic, and only two evaluate bias in the extraction of social constructs from EMRs, but those studies were not then contrasted with such that it wasn't clear what the advantage or continuation of progress over those papers is.

Authors' response 1:

We thank the reviewer for pointing out that we did not adequately contrast our work with the two prior studies that evaluated bias in social construct extraction. We have revised the Introduction to explicitly distinguish our contributions.

Both prior studies extracted social determinants of health (SDoH) from clinical text (employment, housing, etc.) and assessed bias as a secondary outcome. Specifically, Yu et al. developed the SODA package for SDoH extraction and found >16% performance gaps across racial groups, but did not incorporate fairness interventions or systematically evaluate demographic intersections. Guevara et al. extracted SDoH using Flan-T5 models and assessed bias by testing if predictions changed when race/ethnicity descriptors were injected into text, finding that fine-tuned models were less biased than ChatGPT.

Unlike prior work that extracted SDoH with bias as secondary consideration, our study explicitly focuses on classifying patient race from clinical text, with fairness as a central design objective. We implement fairness constraints directly in the loss function (equalized odds-inspired regularization) during model training, rather than only evaluating bias post-hoc. We propose a CNN-based hierarchical model that explicitly captures the sentence-level and word-level structure of clinical documentation, demonstrating that task-specific architectures can outperform transformers in both accuracy and fairness. We address 9-class race classification (vs. binary or few-class tasks in prior work) under extreme class imbalance (76% absent class), with comprehensive fairness metrics including representation ratios, false positive/negative rates across groups, and equalized odds violations. We systematically evaluate bias across demographic intersections (sex x age x race), revealing disparities that emerge at intersectional levels not captured by single-attribute analysis. Finally, we employ a two-phase active learning framework to address the 71:1 race-absent to race-present ratio, substantially improving positive class representation.

We have added this comparison explicitly in the **Introduction** to clarify how our work builds upon and extends beyond these important prior studies:

Despite the urgency of these concerns, only a small number of studies have explicitly evaluated algorithmic bias in the extraction of social constructs from EHRs; to our knowledge, only two have evaluated bias in this context[31, 32]. Yu et al. identified performance gaps exceeding 16% across racial groups when extracting social determinants of health, highlighting persistent disparities by race and gender[31]. Similarly, Guevara et al. assessed algorithmic bias by introducing race and ethnicity descriptors into clinical narratives, demonstrating that fine-tuned models exhibited lower bias than ChatGPT[32]. However, in both cases, bias evaluation was treated as a secondary consideration within broader studies of social determinants extraction, rather than as a central design objective. Moreover, neither incorporated fairness constraints during model training or systematically examined bias across intersecting demographic axes (race, sex, and age). There is a pressing need for more comprehensive assessments of how bias emerges during data collection, model development, and deployment, especially for underrepresented groups, to address bias in predictive modeling[33]. Furthermore, it is equally important to explore and evaluate strategies that can improve model fairness and promote more equitable predictive outcomes.

This study investigates the automatic document-level classification of patient race from unstructured clinical text within EHRs as a critical step toward capturing social constructs that inform care phenotypes, enabling the assessment of healthcare equity and the identification of populations receiving lower-quality care. Unlike named-entity recognition or sentence-level concept extraction, which identify local mentions of race within notes, our objective is to determine each patient's overall racial category based on their complete longitudinal documentation. This formulation aligns with the practical use case of augmenting or validating structured demographic fields and enables population-level analyses of equity and care quality. To achieve this and address the current gaps in the literature, we first implemented an active learning framework to efficiently guide the annotation process by prioritizing notes likely to contain race-related information. We then compared state-of-the-art transformer-based models with a hierarchical convolutional neural network designed to represent the multi-level structure of clinical documentation, thereby capturing contextual dependencies across sentences and encounters. Next, we evaluated the impact of fairness-aware optimization techniques, including a loss function inspired by equalized odds, to determine whether such constraints improved predictive balance across racial groups. Finally, we analyzed potential sources of bias by examining

how dataset composition and sensitive attributes such as sex and age influenced model performance. This work seeks to develop more robust, interpretable, and equitable methods for racial, and more widely social determinants of health, information extraction from clinical text, ultimately supporting the creation of more representative datasets for health equity research

Reviewer #2's comment 2:

2. The BERT-based LLM's perform so poorly on what seems to be a relatively simple tasks (as noted by the hierarchical CNN performing incredibly well) that it causes worries of some problem during training, especially when the fairness aware training largely fixes the problem (and not just at the expense of majority group performance).

Authors' response 2:

We thank the reviewer for highlighting this important observation regarding the performance gap between the fairness-unaware and fairness-aware transformer models. This difference reflects a deliberate aspect of our experimental design, which we recognize should be clarified. Our dataset exhibits extreme class imbalance, with approximately 76% of records lacking race information (the "absent" class) and the remaining 24% distributed across nine racial categories (several of which contain fewer than 100 training samples). This results in a highly imbalanced multi-class classification problem. To establish a realistic baseline, the fairness-unaware models were trained without any balancing techniques (i.e., no weighted loss, resampling, or synthetic oversampling), representing the "naive" approach that is often used in practice when fairness or imbalance considerations are not explicitly addressed.

In contrast, the fairness-aware models incorporated class-weighted loss functions (normalized to prevent instability) alongside fairness-specific regularization using the equalized odds constraint. While class weighting is a common method for imbalanced learning, we considered it part of our fairness intervention because it directly influences performance parity across groups and allows us to assess the combined effect of imbalance-handling and explicit fairness regularization. The poor performance of fairness-unaware transformer models, particularly their near-zero precision and recall for minority groups, is thus expected and consistent with the literature on imbalanced data training. Transformers tend to optimize toward majority-class accuracy in such settings, leading to systematic underperformance on minority classes. Furthermore, most studies publish only aggregate results (micro-averaged scores) in which if you look at those results only in our study for the transformer models without fairness objectives it would appear that they perform quite well. The hierarchical CNN, by contrast, performed better even in the fairness-unaware configuration due to architectural advantages such as its hierarchical attention structure, which captures sparse, sentence-level race mentions, as well as implicit regularization effects from its layer design. Furthermore, it was not constrained by pre-training bias and the model is much simpler in design and its convolutional approach might potentially have made it more efficient in detecting sparse race information.

Finally, the strong gains observed in the fairness-aware transformer models confirm that the combination of class weighting and fairness regularization effectively mitigates imbalance-driven disparities. Importantly, this improvement did not come at the cost of majority-class performance, indicating that the fairness-unaware models were not capacity-limited but rather undertrained on minority groups. To address potential confusion, we have added clarification in the **Methods** section under the **Fairness constraints** subsection.

This experimental design reveals an important finding: models deployed without fairness considerations (including basic imbalance handling) can exhibit severe disparities, even when using state-of-the-art architectures. The dramatic improvement with fairness interventions demonstrates that fairness is not inherently costly, it primarily requires deliberate design choices rather than fundamental performance trade-offs.

Reviewer #2's comment 3:

3. This then mitigates a central point of the discussion of worrying about LLMs detecting and propagating hidden biases as the results seemingly show they can't detect them. Addressing point 2 would fix this, but as it stands that discussion point felt hollow.

Authors' response 3:

We thank the reviewer for this insightful observation. We agree that, as initially written, the discussion may have appeared to overstate concerns about bias propagation, given that the fairness-unaware transformer models failed to detect race information. This failure, however, should not be interpreted as evidence that these models are unbiased or that race cues are absent from clinical narratives. Rather, it reflects a different form of bias, an optimization bias arising from extreme class imbalance and majority-class dominance, whereby the model effectively "learns to ignore" underrepresented groups. In this sense, the inability to detect minority-group signals is itself a manifestation of bias, not neutrality.

This is evident in the fairness-unaware hierarchical CNN that was trained on the same dataset without any balancing or fairness constraints and that successfully identified race-related information across multiple groups, confirming that these cues are indeed present in clinical narratives. The disparity between the transformer and CNN results reflects architectural and optimization biases: transformer models trained under extreme imbalance tend to overfit majority-class patterns and disregard minority signals, whereas the hierarchical CNN's sentence-level structure and implicit regularization allowed it to capture sparse race mentions more effectively. In this sense, the transformers' inability to detect race information is not evidence of neutrality but of a bias toward majority-class representation. We have clarified this point in the revised **Discussion**.

Reviewer #2's comment 4:

4. The Hierarchical CNN does't seem to have 512 max token length (or it does but just per sentence which are all handled differently)? Could this be a source of the performance gap? Additional results even just in the supplement regarding the rate of truncation, analysis to see whether and how often the label information was lost through truncation, performance on truncated vs. non-truncated data points would be valuable.

Authors' response 4:

We appreciate this important observation. As clarified in the revised **Methods** section based on review #1's comment, our corpus consists of short social history and risk factor sections with 98 % of records well below the 512-token limit of transformer models. Thus, truncation was extremely rare and did not affect label availability. Both the transformer and hierarchical CNN architectures therefore had full access to the same document content. The hierarchical CNN's stronger performance reflects architectural design rather than any advantage from input length. We believe the clarification already incorporated into the revised manuscript adequately addresses this concern, so no further changes were made.

Reviewer #2's comment 5:

5. I worry given the small size of the dataset that intersectional biases may be extremely noisy, especially for minority groups given their small prevalence and the number of groups to look for bias in.

Authors' response 5:

We appreciate this important point. We fully agree that intersectional analyses can become unstable when subgroup counts are small, particularly for under-represented populations such as Indigenous or mixed-heritage patients. In our study, intersectional metrics were reported primarily to highlight visibility of these groups rather than to draw strong statistical conclusions. We therefore interpret these results qualitatively and in relation to broader performance patterns. We have clarified this limitation in the revised manuscript (Discussion) to ensure appropriate interpretation of intersectional findings.

We hope that the answers and responses to the reviewer's comments are satisfactory.